Multimodal Hate Video Classification on HateMM

EPFL

Youssef Dib, Kevin Abou Jaoude, Rami Aschkar · **Group 07**

EE-559 Deep Learning · EPFL, Spring 2026

Problem definition

Problem. Hateful content online increasingly travels as video, where meaning is carried jointly by visual context, audio delivery, and the spoken message. Single-modality classifiers struggle on cases where these three signals must be combined to distinguish hate from satire or critical discourse.

Hypothesis. A small transformer trained on top of frozen pretrained per-modality encoders can outperform every unimodal baseline on a real corpus of hateful and non-hateful videos, with gains that are not uniform across target communities.

Dataset

HateMM (Das et al., ICWSM 2023), released under CC BY 4.0 on Zenodo. We use it as-is, without modification or redistribution.

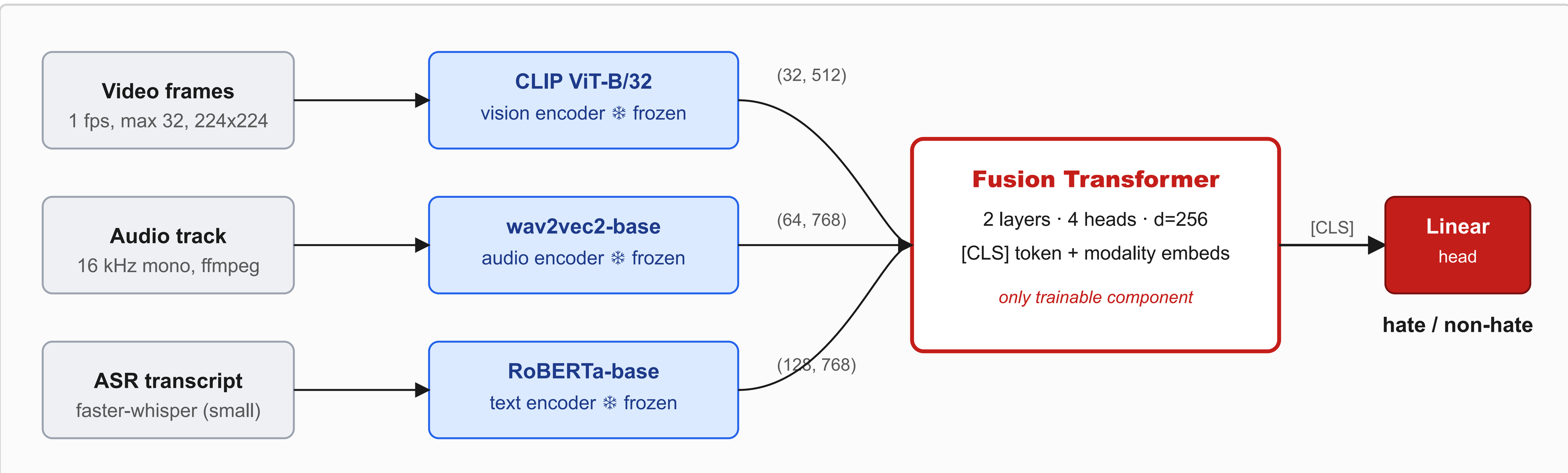
- **1083 BitChute videos** (431 hate, 652 non-hate); ~43.5 hours, 6.3 GB.
- Per-video labels: binary hate flag, target community, hateful-segment timestamps.
- 80/10/10 stratified split (863/109/109), fixed seed; preprocessing succeeded on 1081/1083 videos.

Method

Pipeline in three stages.

1. **Preprocess.** Sample frames at 1 fps capped at 32 per video, center-cropped to 224 x 224; extract 16 kHz mono audio via ffmpeg; transcribe speech with faster-whisper.
2. **Frozen feature extraction.** CLIP ViT-B/32 per frame, wav2vec2-base mean-pooled into 64 audio tokens, RoBERTa-base token embeddings (length 128) on the transcript.
3. **Fusion.** A 2-layer transformer (d = 256, 4 heads, GELU, pre-norm) ingests the concatenated per-modality token sequences plus learnable modality-type embeddings and a [CLS] token; the [CLS] output goes through a linear head for binary classification.

Why frozen encoders. Keeping all three encoders frozen makes the modality ablation a fair comparison: only the small fusion head varies between configurations. Restricting the same architecture to a single modality at the input gives our unimodal baselines.



Related works

- **HateMM [1].** Defines the task; releases the dataset and target-community labels we audit against.
- **CLIP [2].** Provides our frozen vision encoder; we use the image branch only.
- **Multimodal fusion transformers [3].** Motivate the cross-modal-attention design; we adopt a lighter [CLS]-readout variant suited to the dataset scale.

Reproducibility

AdamW (lr 1e-4, batch 16, BCE), early stop on val macro-F1; 3 seeds {42, 1337, 2025} with the split held fixed. Single NVIDIA V100 on EPFL Noto, using the EE-559 course kernel plus *opencv-python-headless* and *faster-whisper*. Full pipeline and aggregator at **github.com/YDIB11/Mini-Project**.

References

1. M. Das, R. Raj, P. Saha, B. Mathew, M. Gupta, and A. Mukherjee, "HateMM: A Multi-Modal Dataset for Hate Video Classification," in *Proc. AAAI ICWSM*, vol. 17, 2023, pp. 1014-1023.
2. A. Radford et al., "Learning Transferable Visual Models From Natural Language Supervision," in *Proc. ICML*, vol. 139, 2021, pp. 8748-8763.
3. A. Nagrani, S. Yang, A. Arnab, A. Jansen, C. Schmid, and C. Sun, "Attention Bottlenecks for Multimodal Fusion," in *Proc. NeurIPS*, vol. 34, 2021, pp. 14200-14213.

Validation

Table 1. Modality ablation on the test split (n = 109), mean ± std over 3 seeds {42, 1337, 2025}.

Modality	Acc	F1	Prec	Rec	AUROC
V + A + T	0.79 ± 0.05	0.78 ± 0.05	0.74 ± 0.08	0.74 ± 0.08	**0.88 ± 0.01**
Text only	0.79 ± 0.02	0.78 ± 0.02	0.75 ± 0.06	0.73 ± 0.08	0.83 ± 0.02
Video only	0.72 ± 0.03	0.71 ± 0.02	0.64 ± 0.06	0.68 ± 0.06	0.81 ± 0.01
Audio only	0.67 ± 0.02	0.67 ± 0.03	0.56 ± 0.02	0.71 ± 0.05	0.77 ± 0.03

Table 2. Per-target-community test accuracy (n ≥ 15), mean ± std over 3 seeds.

Community	n	Fusion	Video	Audio	Text
Blacks	46	**0.75 ± 0.05**	0.70 ± 0.04	0.59 ± 0.06	0.73 ± 0.03
Others	39	0.84 ± 0.01	0.74 ± 0.09	0.68 ± 0.11	**0.85 ± 0.05**
Jews	15	0.73 ± 0.13	0.67 ± 0.07	0.71 ± 0.04	**0.76 ± 0.04**

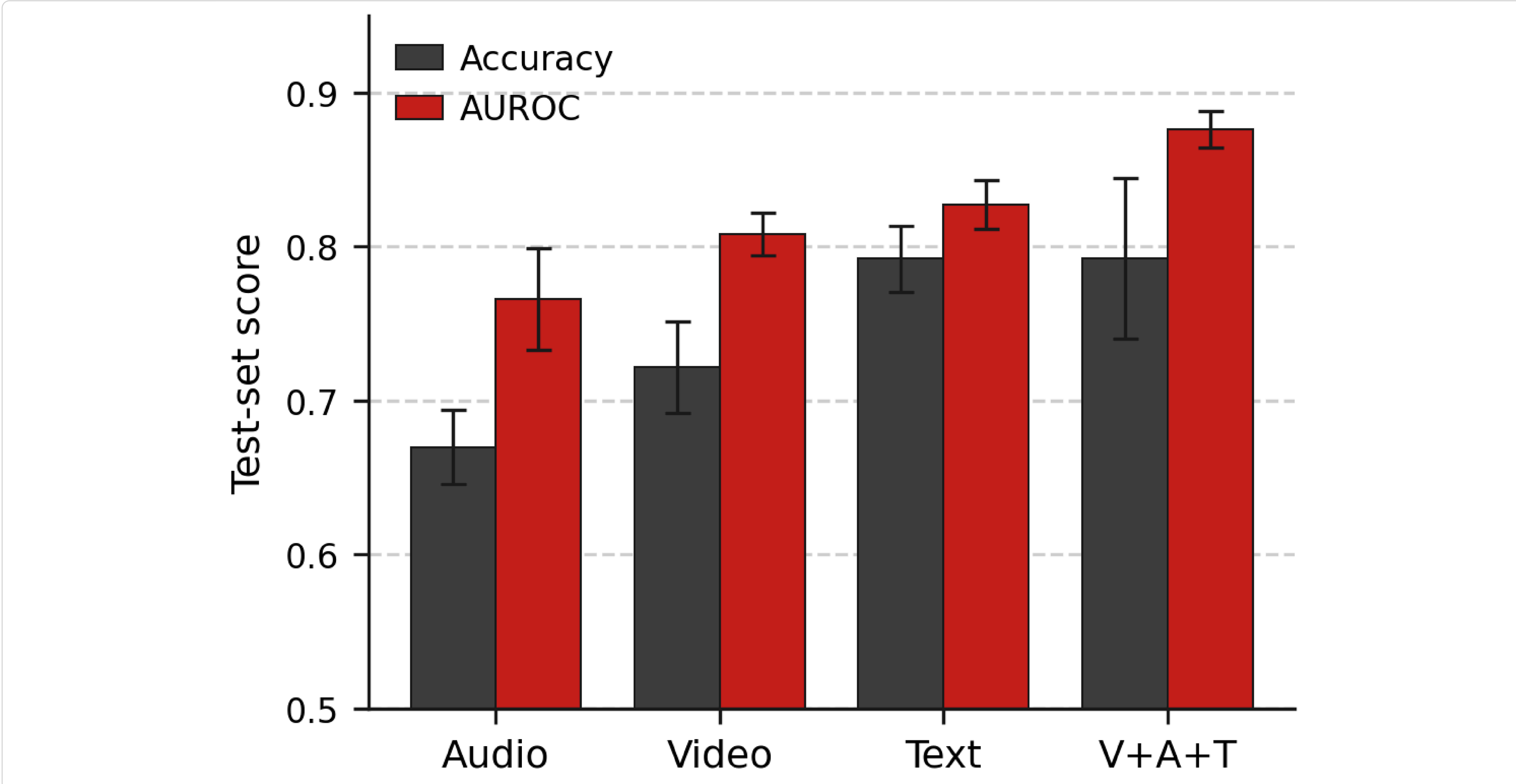


Figure 1. Test accuracy and AUROC across the four models, mean ± std over 3 seeds, sorted ascending by AUROC.

Key observations

Robust ranking gain, noisy threshold gain. Fusion wins AUROC by 5 points over the strongest unimodal (text), with the gap exceeding twice the combined standard deviation. On threshold-dependent metrics (accuracy, F1) fusion matches text on average with markedly higher per-seed variance (std 0.05 vs 0.02).

Text is the strongest unimodal baseline in this dataset, consistent with the verbal nature of much hate content.

Per-community, all fusion-vs-text gaps lie within combined seed variance. On the smallest group *(Jews, n = 15)* the fusion model's seed std (0.13) *exceeds* its mean advantage by an order of magnitude.

Limitations

- **Small test set (n = 109).** Per-community stats for groups with n < 15 are excluded; reported groups still carry appreciable variance.
- **Heuristic community-label normalisation.** Multi-target labels canonicalised to a sorted comma-joined form, conflating intersectional rows with singleton ones.
- **ASR quality varies.** Shouting, music, and accented speech degrade the text modality on a subset.
- **Architectural choices.** Frozen backbones cap absolute performance, and wav2vec2 hidden states are pooled to a fixed 64 tokens (over-averaging long videos).

Conclusion

Fusing three frozen pretrained encoders through a small transformer head **robustly improves AUROC** over every unimodal baseline on HateMM (0.88 ± 0.01 vs 0.83 ± 0.02 for text). On threshold-dependent metrics the gain washes out into seed variance, a finding single-seed reporting would have missed. Per-community, text alone is competitive with or better than fusion on every reported group, so fusion delivers a calibrated ranking gain rather than a uniform threshold-level improvement over a strong text-only baseline.

Practical implication. Fusion is the right choice for ranking-based pipelines (triage, human-in-the-loop review). For a fixed-threshold binary decision on this dataset, text-only is a strong and cheaper alternative.